MILESTONE M0 · EVALUATION BEFORE MODELS

Provenance

An evidence-surfacing system for news claims. This is a record of the work done on milestone M0: the leakage critique that motivates the project, the out-of-distribution corpus built to test it, and the defects found along the way — including the ones I introduced.

2026-09-16 suite v4 · sha 1adb5673 45 commits 341 tests 30 decisions branch master

1. The three things that do not move

Stated in `README.md` before any code existed, and each one is load-bearing rather than aspirational.

- **The system never outputs a truth verdict.** Not a score, not a label, not a hedged one. It surfaces evidence and says where the evidence came from. This is D001, and it is why ablation 4 below reports a prediction distribution instead of an accuracy.
- **Evaluation is written before the model.** The collection procedure was specified in full, and committed, before the collector existed (D013). A procedure designed after the data is visible drifts toward whatever the data happens to contain, and the drift is invisible to the person causing it.
- **Every fetched byte is hostile.** The fetcher validates addresses, pins the connection to the address it validated, revalidates on every redirect, and caps both size and decompression mid-stream.

2. The finding: what a 99% accuracy figure is actually measuring

The standard project on this topic trains a classifier on the ISOT-style corpus and reports accuracy in the high nineties as an ability to detect false news. That corpus's true half is wire copy from one agency; its false half is a heterogeneous scrape. If accuracy survives the removal of information that cannot possibly carry veracity, the classifier was never measuring veracity.

CONDITION	LINEAR, FULL	DISTILBERT, REDUCED	INFORMATION REMOVED
Baseline	0.9849	0.9920 †	none
First sentence removed	0.9756	0.9710 †	leading sentence
Length matched	0.9824	0.9700 †	document length
Structure only	**0.8184**	**0.832** 0.765–0.872, 3 seeds	**every content word**

Structure only is the decisive row. Every content word is replaced with a shape token — `Trump` becomes `Xxxxx`, `2017` becomes `0000` — leaving punctuation, capitalisation and length. Nothing a human could read survives. Accuracy is still 81.8% and 83.5% against a 50% chance baseline. Formatting residue alone carries roughly two thirds of the distance from chance to the headline figure.

> **A claim withdrawn.** An earlier version of this report said the two model families "land 1.7 points apart" on this condition, and that the agreement mattered more than either number. It rested on a single transformer run made before the runner seeded anything — `torch.manual_seed` was never called. With seeding controlled the transformer spans **0.765 to 0.872** across three seeds, and the linear probe's 0.818 sits inside that range. The agreement was one draw, and the claim is withdrawn rather than softened.
>
> What survives is weaker and still sufficient: both families land far above chance with every content word destroyed, at 0.818 and a worst-seed 0.765. That supports the signal being a property of the corpus rather than of one probe. It does not support them agreeing closely.

The linear column is single seed and deterministic, measured not assumed: five seeds return 0.9849 and 0.8184 to four decimal places, because the mirror ships a fixed split, the structure-only transform is deterministic, and `LogisticRegression` under lbfgs ignores `random_state`. **† marks a single unseeded transformer run** — not reproducible, and with a delta smaller than that column's 10.8-point seed spread, so those two rows establish nothing. They stay in the table without a delta, because deleting them would hide that they were measured.

A document's comma placement does not change when its content becomes false. What the classifier separates is two publishing pipelines — one house style guide, one HTML scrape — by their formatting residue.

Two of the plan's own predictions were contradicted

`EVALUATION.md` expected ablations 1–3 each to remove a large fraction of the accuracy. First-sentence removal costs 0.9–2.1 points; length matching 0.3–2.2. Both are nearly nothing, and both are reported as contradicted rather than quietly reinterpreted.

There is a measured cause for the first. The `CITY (Reuters) –` dateline survives in only **0.3%** of wire documents across all three reachable mirrors — they arrive pre-stripped — so ablation 1 is a lower bound, not a clean test of the dateline hypothesis. The leakage turns out to be distributed across the document rather than concentrated in one deletable sentence. That is worse for the corpus, and it is why D010 bars this corpus from augmentation and pretraining use as well as from direct training: the leakage is co-extensive with the labels.

3. The refusal asymmetry: what a web-collected news corpus can contain

This is a second measured result, not a note about why the corpus is short. Sixty-nine publishers were approached under one procedure: robots.txt honoured, one conservative crawl delay, one user agent, refusals recorded and never worked around (D019). The procedure did not vary by publisher. What varied was who answered.

SIDE OF THE PUBLISHER POOL	APPROACHED	YIELDED ANYTHING	RATE
Centre or unrated	11	9	82%
Left of centre	34	22	65%
Right of centre	**24**	**10**	**42%**

Newsmax, OANN, the Daily Wire, The Federalist, National Review and Infowars all returned nothing: HTTP 403, no readable feed, or no sitemap carrying a publication date. **A right-of-centre publisher in this pool was about half as likely to be collectable as a centre one.**

> **This binds every project in this space, not just this one.** Any corpus assembled by fetching news from the open web inherits it. A project that widens its publisher pool until the document count looks respectable has not solved it — it has selected on collectability and published the result as if it described the press.

Collectability is not a neutral property. It tracks feed and sitemap hygiene, CDN and bot-management policy, and whether an outlet runs a hard paywall — all facts about publisher infrastructure, and none of them independent of the kind of publisher an outlet is. So the sampling frame of a web-collected news corpus is correlated with exactly the axis such corpora are usually used to analyse.

That is the same category of error as the leakage in section 2, arriving one level earlier. In section 2 the labels carried publisher identity. Here the *sampling frame* does, before a single label is applied. A model trained on such a corpus would learn a property of who answers an HTTP request, and its accuracy would not reveal that, because the corpus would look better for it — a corpus with fewer refusals is larger, more balanced on paper, and wrong in a way nothing downstream can detect.

What this project did about it. Nothing, deliberately, beyond measuring and reporting it (D028). The pool was not widened, refusals were not retried, and the user agent was never changed to get past a 403. The corpus is short as a direct result, and that is the correct trade: the alternative buys documents by importing a bias the project exists to document.

What the measurement does not establish. Why the asymmetry exists. Bot-management vendor choice, budget, CMS age and deliberate policy are all consistent with it, and this project has no evidence separating them. It also says nothing about publishers outside this 69-domain pool, which was itself built from Media Bias/Fact Check's coverage. The finding is that the asymmetry is there and is large, not that its cause is known.

4. Ablation 4: what the model says about 2026 news

> **No accuracy appears in this section, and none is available.** Accuracy requires veracity labels, and the out-of-distribution suite deliberately carries none (D001). Each figure below is the share of a group that the ISOT model assigns to the corpus's wire-service half — the class that corpus treats as real. **No group here is established as more or less truthful than any other.**

The plan expected accuracy decaying toward chance on later text. That number cannot exist, so what is reported instead is what the model *says*. Both models were applied unchanged to `ood_suite_2026_v4` : 266 documents, 41 publishers, nine months. These supersede figures measured on the first freeze, which was 384 documents with 289 published on a single day; both runs are kept in `transfer.json`, keyed by corpus hash.

GROUP	DOCUMENTS	BASELINE	STRUCTURE ONLY
Whole suite	**266**	**22.2%**	**56.4%**
Wire	30	36.7%	90.0%
Regional	27	33.3%	37.0%
National	73	17.8%	68.5%
Partisan	136	19.1%	46.3%
Lean left	68	27.9%	39.7%
Lean right	50	20.0%	54.0%
Flagged low credibility	18	16.7%	44.4%

A model scoring 0.9849 on its own held-out split calls **78% of real 2026 news false** — including 63% of actual wire copy, the half of its training data it is supposed to recognise. On the superseded one-day suite that was 77.9%; across nine months and 41 publishers it is 77.8%. **This is the result the report rests on, and it is the one that replicated.**

It also survives resampling. Publishers are the sampling unit, not documents — documents from one outlet share a house style and a topic mix, and DATA.md makes the domain the unit everywhere else for that reason. Resampling the 41 publishers with replacement over 2000 draws puts the whole-suite share at **22.2%, 95% interval [14.7%, 30.2%]**, well clear of 50%. Refitting the model changes nothing: five seeds give 0.2218 every time.

The lean gap did not replicate, and is withdrawn

The first freeze showed 36.8% of lean-left documents called real against 10.5% of lean-right — a 26.3 point gap, reported as a finding. It does not survive.

MEASURE	FIRST FREEZE	V4	V4, 95% OVER PUBLISHERS
Baseline, lean left	36.8%	27.9%	[13.6, 42.6]
Baseline, lean right	10.5%	20.0%	[6.6, 48.3]
Baseline gap	**+26.3 pp**	**+7.9 pp**	**[−25.5, +27.8]**
Structure-only gap	**+10.5 pp**	**−14.3 pp**	**[−36.9, +8.0]**

Both gaps straddle zero. Neither is established as non-zero once the publisher is the sampling unit. The intervals are wide because `lean_right` holds six outlets, and that width is the honest width — a narrower one would come

from treating 50 documents from 6 publishers as 50 independent draws.

The obvious explanation was tested and does not hold either. If the September 11 concentration produced the original gap, September inside v4 should still show it. It does not: **+4.5 pp in September against +3.6 pp for January–August**, a difference this corpus cannot resolve. Two suites differed in dates *and* in publisher mix at once; this rules out the first, not the second, and the cause is left unexplained rather than fitted to the nearest plausible story.

Two further things bear on reading the pooled number. The pooled gap is *larger* than either period's — Simpson's paradox, since every rate is higher in September while lean-left is 54% September against lean-right's 36%. And September's elevated rates are confounded with collection path: September is the live-window month, the other eight came from archive backfill, and this suite cannot separate them.

> **The flagged row is not the model working.** Low credibility is Media Bias/Fact Check's label, not a truth verdict, and in this corpus it sits entirely on right-of-centre publishers (D017), so it cannot be separated from the lean effect above it. The model has never seen a truth label for any of these articles.

5. The out-of-distribution corpus

The suite that produces the honest number: real articles published after the training snapshot of 2026-01-01, stratified by publisher type, carrying no veracity labels by design.

The first freeze failed its own plan, and was reported as failing it

384 documents, 31 publishers — and **289 of them published on a single day**, the 25th anniversary of the September 11 attacks, with 27% of the corpus mentioning the attacks. Feeds and news sitemaps list only the last day or two, so one collection run cannot spread across weeks as the plan requires. The freeze was recorded in the manifest and written up as inadequate rather than presented as a result.

Two fixes, in sequence

D022 — reach past the live window. Archive sitemaps list far more than feeds, but date entries only by last modification, and a page edited this year may have been published years ago. D019 refuses `lastmod` as a publication date, so each archive candidate is fetched and kept only if the page itself states a publication date after the snapshot. Most do not, which is what the per-publisher fetch budget is for.

D024 — the cap was on the wrong half. The first backfill run improved the spread but still left 459 of 528 documents in September. The per-month cap governed only the archive; the live window, which is itself one news cycle, was uncapped. One cap now covers a publisher's whole sample, and the live window was cut from 15 articles to 3.

What the rebuild produced

MEASURE	FIRST FREEZE	FROZEN V4	PLAN
Months represented	1	9	spread across weeks, not one cycle
Largest single day	75%	20.7%	no single news cycle dominates
Mentioning 9/11	27%	0.8%	—
Publishers	31	41	—
Wire documents	15	30	thinnest stratum, still thin
Documents with identity residue	1	0	from 42 before the rule fixes (D026)
Non-articles excluded by rule	—	2	newsletter issue, donation page (D027)
Documents	**384**	**266**	**400–600, floor 300**

Reuters is the clearest gain. Its homepage and feeds are disallowed by `robots.txt`, so it contributed nothing before; its archive paths are permitted, so it now supplies three articles per month across nine months — a publisher previously scored as a total refusal.

Why 266 and not 400

Collection returned 331 documents; balancing the partisan stratum by lean (D017) dropped 59. The reason the balance costs that much is the refusal asymmetry in section 3, which is a finding in its own right rather than an explanation for a shortfall. The count consequence is in section 11.

The audit, and a process error of mine

The collection plan makes a clean audit and a residual check preconditions of freezing. I said so, then wrote a job that ran both and froze regardless of what they reported. The audit flagged publisher-identity residue in **42 of 331 documents**; the freeze proceeded anyway. That freeze (v2) was superseded rather than kept.

Re-running the existing rules removed nothing. Reading all 42 by hand found two different problems wearing one label:

- **Real residue the rules missed.** Links back into the publisher's own domains — hosts are lowercase and unspaced, so a case-sensitive name mask never sees `bbc.co.uk/learningenglish`. Institutional bylines with no surname to match. Standalone captions with no credit word. A licence notice inside a sentence, where the line-anchored rule could not reach. And a masking gap: `outlet_names.yaml` records `the Post` lowercase, so every sentence opening "The Post's broader goal" went unmasked.
- **The audit flagging prose.** "By Friday afternoon, Hezbollah had issued a statement" is not a byline. "Advanced Placement (AP)" is not the Associated Press. "in the Atlantic Ocean" is not The Atlantic. Its self-link check flagged every third-party link an article cited, because a pattern can see that a URL is present but not whose it is.

Both sides were fixed in one pass, each new rule tested against the prose it must not touch as well as the residue it must remove. **Flagged documents fell from 42 to 2**, and both survivors are documents that are not articles — a newsletter issue and a donation page. No text rule fixes that; it is a selection failure, and the audit now names the two kinds separately (D026).

> **None of it moved the thing underneath.** Predicting the stratum from text, grouped by publisher, scores 0.5981 against a 0.5988 majority baseline — strata are not separable, which is the result the plan wants. But publisher identity is still recoverable at **38.3% against an 8.2% chance baseline** across 41 classes, and 28.3% from structure alone. That is where it sat before the rule work. Removing explicit markers did not touch the stylistic fingerprint underneath, and marker stripping has reached its limit here.

6. Defects found by running the thing

Each was caught by a test failing or a real fetch misbehaving. None would have been found by reading the code, and several are mine.

robots.txt was under-blocking

Associated Press publish `Disallow: /*.rss`. Python's `urllib.robotparser` has no wildcard support and reports that path as *allowed* — a publisher stated a restriction and the parser granted permission. Rule matching is now implemented directly: anchored regexes with metacharacters escaped, longest match wins, agent-specific groups replacing the wildcard group. A test asserts the standard library still gets it wrong, so the reason for the custom matcher cannot be forgotten.

Two address ranges the standard library misses

`100.64.0.1` (carrier-grade NAT) reports `is_private = False`. `64:ff9b::7f00:1` — NAT64 wrapping `127.0.0.1` — reports `is_global = True`. Both are live SSRF bypasses against a naive check. Every IPv6 form embedding an IPv4 address is now unwrapped and re-checked.

A size cap that could never fire

httpx only counts wire bytes on a genuinely streamed response. My test fixture was pre-read, so the counter sat at zero and the test passed while exercising nothing. Fixtures now stream, and the test additionally asserts the abort happens part-way through the body.

The stratum had become a proxy for politics

Routing publishers by credibility left `partisan` at 15 left-of-centre to 4 right, and the quality stratum at 7 right to 0 left. A model separating strata would have learned politics while appearing to learn quality — the same category error this project exists to document. Quality became an orthogonal flag (D017) and `partisan` moved to 23:20 on identical candidates.

Seventeen suppressed type checks hiding one real risk

Collectors took `robots: RobotsCache`, a concrete class holding a live HTTP client, so every test double was passed under `# type: ignore[arg-type]`. The suppressed check is the one that catches a double missing `wait_for` — the call that paces every request. Three `runtime_checkable` protocols replace it, split by what callers actually use. A single four-method protocol was written first and the new conformance test failed it: the demo's double has no `sitemaps_for` because the demo never calls it. The interface was wrong, not the test (D025).

Two commits landed on a red suite

Both times because `pytest | tail` and `mypy | tail -1` report the exit code of `tail`, not of the checker. Verification now reads exit codes directly, with no pipe. In the first case the failing test was correct and the rule it tested was wrong, so the rule was fixed — a failing test is never adjusted to make a suite pass.

7. Governance: what was promised, and what is actually in place

Policy text is not a control. `data/MANIFEST.yaml` records the real state of each obligation, including the ones not met.

OBLIGATION	STATE
Written legitimate-interests balancing test — purpose, necessity, balancing, safeguards, reversing conditions (`docs/balancing-test.md`)	closed
Deletion that actually deletes — refuses targets outside the data root, resolves paths before the containment check, receipt carries a SHA-256 per file and never text	closed
No article text in logs, errors or git	closed, tested
No verdict about any person or claim	closed, structural
Publisher markers stripped before storage, one pipeline for every stratum	closed
Encrypted storage in an EU region — FileVault verified by `fdesetup status` , 2026-09-16; host in the EU	closed, evidenced
Written objection procedure — receipt, proportionate verification, action, logging (`docs/objection-procedure.md`)	closed
A *reachable* contact — the user agent publishes `https://example.invalid/bot` , which cannot resolve, so no publisher contacted during collection had a route either	**open, stated**

The balancing test records what makes the balance uncomfortable as well as what supports it: incidental Article 9 data that no filter can catch in advance, the gap between "published" and "expects to be in a corpus", and the absence of any mechanism by which a data subject could object. It concludes that the interest carries, narrowly, and only because the processing is small, non-decisional, non-redistributed and time-limited.

8. Running it

A demonstration CLI runs one URL through every stage and prints what each did. It bypasses nothing: a URL that `robots.txt` disallows is refused exactly as it is in collection, and there is deliberately no flag to override that.

```
PYTHONPATH=. .venv/bin/python -m pipeline.demo <url>

URL         https://www.theguardian.com/artanddesign/2026/sep/14/...
robots.txt  fetched=True crawl_delay=2.0s sitemaps=2 (137 ms)
fetch       HTTP 200 text/html 386228 bytes (88 ms)
extract     13250 characters (53 ms)
            page date: 2026-09-14  language: en  english ratio: 0.29
            invisible characters: False
            sha256: 560f61d66ea3d613...
stripped    nothing matched the publisher marker rules
```

Exit codes separate refusal (2) from failure (1), so a disallowed URL is never confused with an unreachable one. `--json` gives machine-readable output, `--outlet-name` masks a publisher's self-references, `--verbose` logs each network step.

9. Verification state

CHECK	RESULT
pytest	`341 passed, 2 deselected · exit 0`
ruff check	`All checks passed · exit 0`
ruff format --check	`75 files already formatted · exit 0`
mypy strict (pipeline, eval)	`no issues in 36 source files · exit 0`
pre-commit --all-files	`ruff, ruff-format, gitleaks all Passed · exit 0`

Every exit code above was read directly rather than through a pipe, for the reason given in section 6. 5,445 lines of source against 3,564 lines of tests. `mypy tests/` reports 66 errors — mostly dict invariance in test helpers — and tests are not yet in mypy's `files`; recorded in D025 rather than fixed silently.

10. Decisions

Each entry in `context/DECISIONS.md` carries the alternatives it rejected, the reason each lost, the costs being accepted rather than solved, and the condition that would reverse it.

ID	DECISION
D001	The system does not output a truth verdict
D002	Rhetoric detection is multilabel and span scoped, not document binary
D003	Retrieve then classify stance, rather than end to end
D004	LLM used for claim extraction only
D005	ONNX CPU inference, no GPU in production
D006	Bluesky and Reddit for the circulation signal
D007	Source registry stores observations, never scores
D008	Asynchronous URL fetch, never inline
D009	Frozen holdout inspected exactly twice
D010	The ISOT style corpus is never used for training
D011	Mypy strict covers `eval/` as well as `app/` and `pipeline/`
D012	`data/MANIFEST.yaml` is where corpus use restrictions are enforced
D013	The OOD collection is specified in writing before it is implemented
D014	The transformer reproduction is run at reduced scale
D015	The fetcher pins the connection to the address it validated
D016	Publisher lean is recorded as its own axis, and the stratum confound is reported rather than absorbed
D017	`low_quality` is an orthogonal flag, not a stratum
D018	A publisher's own name is masked in its own articles
D019	Refusals end collection from a host, and `lastmod` is not a publication date
D020	The OOD suite is gated to English at the listing and at the page
D021	Text-only construction rules may be re-applied to a stored run
D022	OOD collection must reach beyond the live publishing window
D023	The aggregator stratum is left empty and reported
D024	The per month cap covers the publisher's whole sample, not the archive alone
D025	Collectors depend on a robots interface, not on the robots cache
D026	Publisher stripping is widened by audit, and the audit is corrected with it
D027	A document that is not an article is excluded by rule, before selection
D028	The corpus stays at 266 and the publisher pool is not widened
D029	The topic requirement is measured rather than dropped, and it holds
D030	The storage control is closed on evidence; the objection route is written but unreachable

11. The topic requirement, finally checked

DATA.md asks for topics spread across politics, health, science and business so that no single news cycle dominates. The freeze reads topic from the article URL path, which resolves for 17 of 266 documents, so the requirement went unverified for the whole of M0. Measured by zero-shot entailment over the text with the MNLI model ARCHITECTURE.md already names for M3, pinned at revision `6f5cf0a2`.

TOPIC	DOCUMENTS	SHARE OF NAMED
politics	80	37.0%
business	75	34.7%
science	33	15.3%
health	28	13.0%
unclear	50	18.8% of all

It holds at the corpus level, and that was not the expected result. All four topics are present and the largest is 37.0% of named documents against the 25% an even split would give. A corpus of 2026 news from a partisan-heavy pool was expected to be dominated by politics and is not. Coverage rises from 6.4% to 81.2%.

It does not hold per cell, and the cells are reported rather than fixed. Wire is 73% business, which is what a wire service's dated archive contains; partisan is 48% politics. The suite is frozen, and rebalancing after seeing the distribution would be selection made with the data visible.

Neither method is a ground truth. Where a path label and an inferred label both exist they agree on 12 of 17 documents. That is 17 documents, so the rate is itself uncertain; the honest reading is that both signals are noisy. The inferred label is a separate field and never replaces the publisher's own filing decision.

12. What is not done

Requires a decision that is not mine to make

- **The corpus is 266 documents against a 400–600 target and a 300 floor.** This is the count consequence of the refusal asymmetry in section 3, not a separate problem. Because the partisan stratum is balanced by lean, the thinner side sets the quota, so collecting more of everything else adds nothing. Getting past 300 means accepting an unbalanced partisan stratum, widening the pool with outlets chosen for being collectable, or accepting a corpus below the floor. The second is rejected in D028 for the reason section 3 gives; the first and third are live and are recorded rather than absorbed.
- **Publisher identity remains recoverable at 38.3% against 8.2% chance.** Marker stripping has reached its limit. Whether that matters depends on what is built on this corpus next.
- **The cause of the original 26.3 point lean gap is not established.** It did not replicate, and the news-cycle explanation was tested and rejected. Dates and publisher mix changed between the suites at once; only the first was ruled out.

Known and outstanding

- The transformer's baseline, first-sentence and length-matched runs are single unseeded draws and are not reproducible. Only structure-only was re-run under seed control.
- The aggregator stratum is empty. Five aggregators were candidates; all five either refuse this collector or publish nothing it can read (D023).
- The wire stratum reaches only two of the three splits, and is 30 documents from 2 publishers.
- Topic per cell is uneven and reported rather than rebalanced: wire is 73% business, partisan 48% politics (D029).
- `tests/` is not covered by mypy; that run reports 66 errors, unrelated to the robots work that exposed it.
- M1 is partially done — fetcher, robots, extraction, demo CLI, Postgres and Redis all exist; the Flask skeleton does not.

> **On the numbers in this document.** Every figure was read from `eval/leakage/results/*.json`, the collection run's own output, or the frozen suite on disk. The DistilBERT column is a reduced-scale run — 6,000 documents at 128 tokens, forced by host load and recorded as D014 — so its absolute values are not a full-corpus reproduction, and only the structure-only gap is large enough to survive single-seed variance. Both leakage columns are single-seed with no confidence intervals, deliberately matching the defaults of the approach under criticism.

Frozen suite `ood_suite_2026_v4`, 266 records, 41 publishers, sha256 `1adb5673758262806c14d5edec3c8b02dd6a338276132734bb0f5d0dd3322415`, frozen 2026-09-16. Supersedes `ood_suite_2026_spread` (v3, sha `f7b361e5`) and the 2026-09-11 freeze of 384 records from a single day. Decisions D001–D030. Corpus text is not redistributable and is not checked in. Regenerate this document from `docs/m0-report.html`.